Variational Inference, Autoencoders, and Applications

Kellen Kanarios

Bayesian Inference

Prior: $\mathrm{p}(\mathrm{z})$

Likelihood: $\mathrm{p}(\mathrm{x})$

Conditional: $\mathrm{p}(\mathrm{x} \mid \mathbf{z})$

Posterior: $\mathrm{p}(\mathrm{z} \mid \mathbf{x})$

Given

Intractable

Bayes rule:

$$\mathrm{p}(\mathrm{z} \mid \mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$$

Bayesian Inference

Prior: $\mathrm{p}(\mathrm{z})$

Conditional: $\mathrm{p}(\mathrm{x} \mid \mathbf{z})$

Given

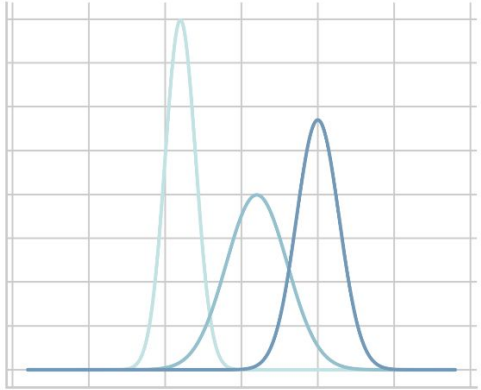
Likelihood: $\mathrm{p}(\mathrm{x})$

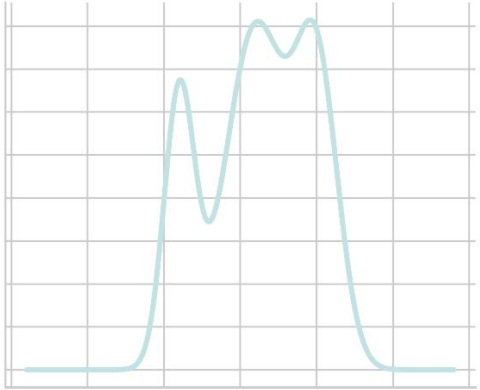
Posterior: $\mathrm{p}(\mathrm{z} \mid \mathbf{x})$

Intractable

Bayes rule:

$$\mathrm{p}(\mathrm{z} \mid \mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$$





Bayesian Inference

Prior: $\mathrm{p}(\mathrm{z})$

Likelihood: $\mathrm{p}(\mathrm{x})$

Conditional: $\mathrm{p}(\mathrm{x} \mid \mathbf{z})$

Posterior: $\mathrm{p}(\mathrm{z} \mid \mathbf{x})$

Given

Intractable

Bayes rule:

$$\mathrm{p}(\mathrm{z} \mid \mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$$

Goal: Want to sample from $\mathrm{p}(\mathrm{z} \mid \mathbf{x}) \propto p(\mathbf{x} \mid \mathbf{z})p(\mathbf{z})$ $\Longrightarrow$ MCMC

Example: Thompson sampling $\mathrm{z} = \begin{bmatrix} \mu_1 & \sigma_1 \\ \vdots & \vdots \\ \mu_n & \sigma_n \end{bmatrix}$

Variational Inference

Variational inference: *Can we formulate desired function / distribution as solution of an optimization problem?*

Variational Inference

Variational inference: *Can we formulate desired function / distribution as solution of an optimization problem?*

$$\mathrm{q}^*(\mathbf{z}) = \arg\min_{q(\mathbf{z}) \in Q} \mathrm{KL}(q(\mathbf{z}) \| p(\mathbf{z} \,|\, \mathbf{x}))$$

Remark: For now assume we find $\mathrm{q}^*(\mathbf{z})$ for every x

Variational Inference

Variational inference: *Can we formulate desired function / distribution as solution of an optimization problem?*

$$\mathrm{q}^*(\mathbf{z}) = \arg\min_{q(\mathbf{z}) \in Q} \mathrm{KL}(q(\mathbf{z}) \| p(\mathbf{z} \,|\, \mathbf{x}))$$

Remark: For now assume we find $\mathrm{q}^*(\mathbf{z})$ for every x

Problem: We cannot compute $\mathrm{p}(\mathrm{z} \mid \mathbf{x})$!!

Variational Inference

Variational inference: *Can we formulate desired function / distribution as solution of an optimization problem?*

$$\mathrm{q}^*(\mathbf{z}) = \arg\min_{q(\mathbf{z})\in Q} \mathrm{KL}(q(\mathbf{z})\|p(\mathbf{z}\,|\,\mathbf{x}))$$

Remark: For now assume we find $\mathrm{q}^*(\mathbf{z})$ for every x

Problem: We cannot compute $\mathrm{p}(\mathrm{z}\mid\mathbf{x})$!!

Enter the ***Evidence Lower Bound (ELBO)***

Evidence Lower Bound

$$
\begin{aligned}
\mathrm{KL}(\mathrm{q(z)} \mid\mid p(\mathbf{z} \mid \mathbf{x})) &= \mathbb{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathrm{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})p(\mathbf{x})}{p(\mathbf{z},\mathbf{x})}\right] \\
&= \mathrm{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})}{p(\mathbf{z},\mathbf{x})}\right] + \mathbb{E}_{q(\mathbf{z})}\left[\log p(\mathbf{x})\right] \\
&= \mathrm{KL}(\mathrm{q(z)} \mid\mid p(\mathbf{z},\mathbf{x})) + \underbrace{\log p(\mathbf{x})}_{\text{Independent of } \mathbf{z}}
\end{aligned}
$$

Evidence Lower Bound

$$
\begin{aligned}
\mathrm{KL}(\mathrm{q}(\mathrm{z}) \mid\mid p(\mathbf{z} \mid \mathbf{x})) &= \mathbb{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathrm{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})p(\mathbf{x})}{p(\mathbf{z},\mathbf{x})}\right] \\
&= \mathrm{E}_{q(\mathbf{z})}\left[\log \frac{q(\mathbf{z})}{p(\mathbf{z},\mathbf{x})}\right] + \mathbb{E}_{q(\mathbf{z})}\left[\log p(\mathbf{x})\right] \\
&= \mathrm{KL}(\mathrm{q}(\mathrm{z}) \mid\mid p(\mathbf{z},\mathbf{x})) + \underbrace{\log p(\mathbf{x})}_{\text{Independent of } \mathbf{z}}
\end{aligned}
$$

$\mathrm{ELBO} = \text{-}\mathrm{KL}(\mathrm{q}(\mathrm{z}) \mid\mid p(\mathbf{z},\mathbf{x}))$ ⟹ Maximize over $\mathrm{q}(\mathrm{z})$

Evidence Lower Bound

$$\text{ELBO} = \text{-KL}(\text{q}(\text{z}) \mid\mid p(\mathbf{z}, \mathbf{x}))$$

$$= \underbrace{\mathrm{E}_{q(\mathbf{z})}\left[\log p(\mathbf{x} \mid \mathbf{z})\right]}_{\text{Maximize likelihood of data}} - \underbrace{\mathrm{KL}(q(\mathbf{z}) \mid\mid p(\mathbf{z}))}_{\text{Regularize to prior}}$$

Distribution of Z typically **smoother / well-behaved** due to KL regularization to nice prior

Beyond Variational Inference: Likelihood Estimation

Want to find $\mathrm{p}(\mathrm{x})$

Z is a **latent** variable

$$\mathrm{X} = \{\mathrm{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\} \sim p(\mathbf{x})$$

Beyond Variational Inference: Likelihood Estimation

Consider:

$$\mathrm{p}_\theta(\mathbf{x}) = p_\theta(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})$$

Want to maximize **log-likelihood** i.e. $\mathrm{E}_{\mathbf{x}\sim p(\mathbf{x})}\left[\log p_\theta(\mathbf{x})\right]$

Beyond Variational Inference: Likelihood Estimation

Consider:

$$\mathrm{p}_\theta(\mathbf{x}) = p_\theta(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})$$

Want to maximize **log-likelihood** i.e. $\mathrm{E}_{\mathbf{x}\sim p(\mathbf{x})}\left[\log p_\theta(\mathbf{x})\right]$

$\mathrm{p}_\theta(\mathbf{x})$ is referred to as the ***evidence*** of how good our model is

Bayes rule:

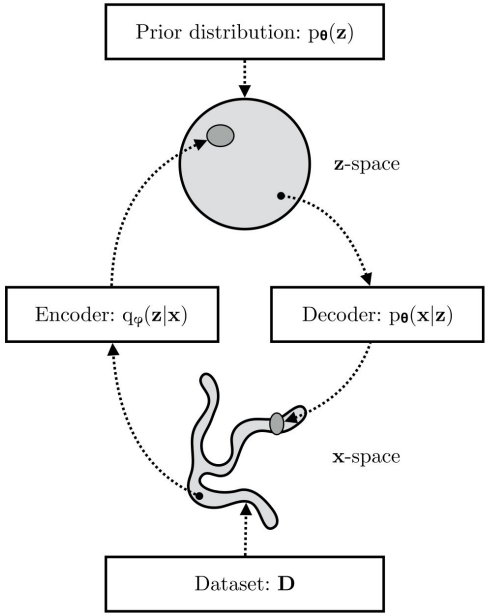
$$\mathrm{p}(\mathrm{z} \mid \mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{x})}$$

$$\Longrightarrow \quad \mathrm{p}(\mathrm{x}) \iff p(\mathbf{z} \mid \mathbf{x})$$

Beyond Variational Inference: Likelihood Estimation

Amortized variational inference *use* ϕ, $q_\phi(\mathbf{z} \mid \mathbf{x})$ *to approximate for all* X

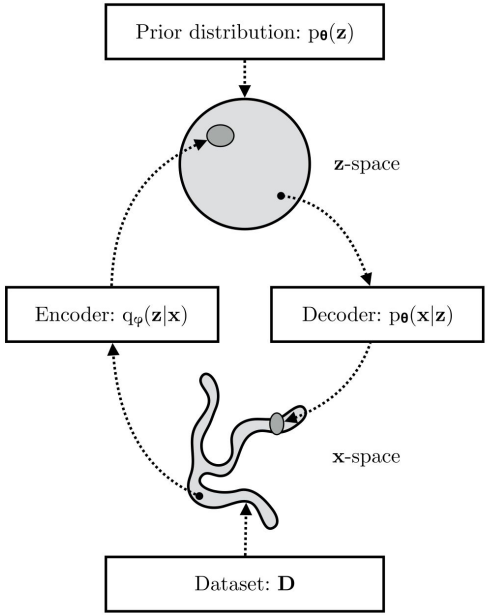
Assume $p(z \mid \mathbf{x})$ is in our function class $q_\phi(\mathbf{z} \mid \mathbf{x})$



Beyond Variational Inference: Likelihood Estimation

Amortized variational inference *use* ϕ, $\mathrm{q}_\phi(\mathbf{z} \mid \mathbf{x})$ *to approximate for all* X

Assume $\mathrm{p}(\mathrm{z} \mid \mathbf{x})$ is in our function class $\mathrm{q}_\phi(\mathbf{z} \mid \mathbf{x})$



Intractable

$$\ell_\theta(\mathbf{x}) = \log \int p(\mathbf{x}, \mathbf{z})\, \mathrm{d}\mathbf{z}$$

ELBO Revisited

$$\log p_\theta(x) = \mathrm{ELBO}(q_\phi(\mathbf{z} \mid \mathbf{x})) + \underbrace{\mathrm{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \| p_\theta(\mathbf{z} \mid \mathbf{x}))}_{\geq 0}$$

$$\Rightarrow \quad \mathrm{ELBO}(\mathrm{q}_\phi(\mathbf{z} \mid \mathbf{x})) \leq \log p_\theta(x)$$

ELBO Revisited

$$\log p_\theta(x) = \mathrm{ELBO}(q_\phi(\mathbf{z} \mid \mathbf{x})) + \underbrace{\mathrm{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \| p_\theta(\mathbf{z} \mid \mathbf{x}))}_{\geq 0}$$

$$\Rightarrow \quad \mathrm{ELBO}(\mathrm{q}_\phi(\mathbf{z} \mid \mathbf{x})) \leq \log p_\theta(x)$$

Equality **if and only if** $\mathrm{q}_\phi(\mathbf{z} \mid \mathbf{x}) = p_\theta(\mathbf{z} \mid \mathbf{x})$

$$\mathrm{q}_{\phi^*}(\mathbf{z} \mid \mathbf{x}) = p_\theta(\mathbf{z} \mid \mathbf{x})$$

Optimizing ELBO

$$\begin{aligned}\mathrm{ELBO}(\phi, \theta) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\ &= \mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}, \mathbf{z})\right] - \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]\end{aligned}$$

Optimizing ELBO

$$\begin{aligned}\mathrm{ELBO}(\phi,\theta) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\ &= \mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x},\mathbf{z})\right] - \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]\end{aligned}$$

$$\begin{aligned}\nabla_\theta \mathrm{ELBO}(\theta,\phi) &= \nabla_\theta \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x},\mathbf{z})\right] \\ &= \mathrm{E}_{q(\mathbf{z}|\mathbf{x})}\left[\nabla_\theta \log p_\theta(\mathbf{x},\mathbf{z})\right] \\ &\approx \nabla_\theta \log p_\theta(\mathbf{x},\mathbf{z})\end{aligned}$$

Optimizing ELBO

$$\begin{aligned}\mathrm{ELBO}(\phi,\theta) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\ &= \mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x},\mathbf{z})\right] - \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]\end{aligned}$$

$$\nabla_\phi \mathrm{ELBO}(\theta,\phi) = \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x},\mathbf{z})\right] - \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]$$

Optimizing ELBO

$$\mathrm{ELBO}(\phi, \theta) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]$$

$$= \mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}, \mathbf{z})\right] - \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]$$

$$\nabla_\phi \mathrm{ELBO}(\theta, \phi) = \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}, \mathbf{z})\right] - \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right]$$

Cannot move inside expectation

Reparameterization Trick

Suppose for some **invertible** g and RV $\epsilon \sim p(\epsilon)$ that $\mathrm{z} = \mathrm{g}(\epsilon, \phi, \mathbf{x})$

Then **for any** f ,

$$\mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[f(\mathbf{z})] = \mathbb{E}_{p(\epsilon)}[f(g(\epsilon, \phi, \mathbf{x}))] = \mathbb{E}_{p(\epsilon)}[f(\mathbf{z})]$$

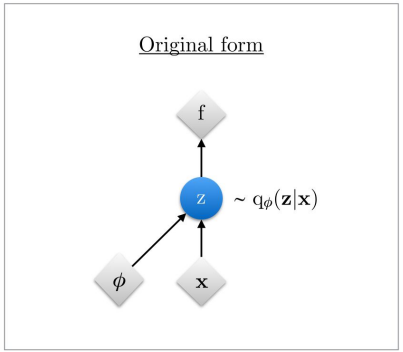
Reparameterization Trick

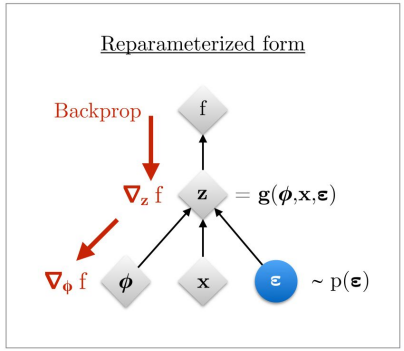
Suppose for some **invertible** g and RV $\epsilon \sim p(\epsilon)$ that $\mathrm{z} = \mathrm{g}(\epsilon, \phi, \mathbf{x})$

Then **for any** f ,

$$\mathrm{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[f(\mathbf{z})] = \mathbb{E}_{p(\epsilon)}[f(g(\epsilon, \phi, \mathbf{x}))] = \mathbb{E}_{p(\epsilon)}[f(\mathbf{z})]$$

$$\implies \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[f(\mathbf{z})] \approx \nabla_\phi f(\mathbf{z})$$





Reparameterization Trick

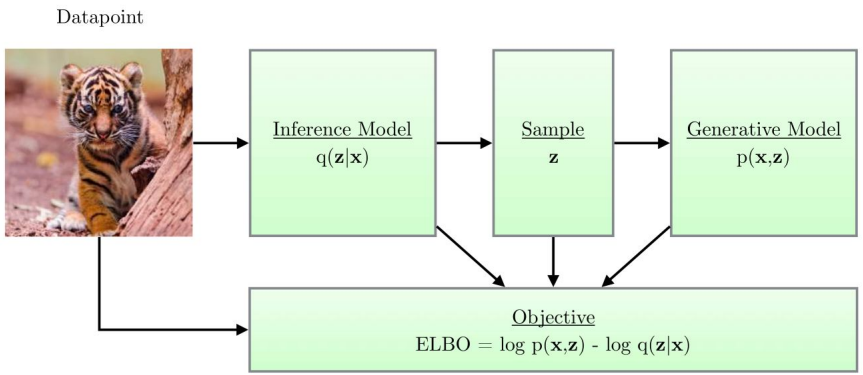
$$\begin{aligned}\nabla_\phi \mathrm{ELBO}(\theta, \phi) &= \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}, \mathbf{z})\right] - \nabla_\phi \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right] \\ &= \nabla_\phi \mathbb{E}_{p(\epsilon)}\left[\log p_\theta(\mathbf{x}, \mathbf{z})\right] - \nabla_\phi \mathbb{E}_{p(\epsilon)}\left[\log q_\phi(\mathbf{z} \mid \mathbf{x})\right] \\ &\approx \nabla_\phi \left[\log p_\theta(\mathbf{x}, \mathbf{z}) - \log q_\phi(\mathbf{z} \mid \mathbf{x})\right]\end{aligned}$$

Where $\mathrm{z} = \mathrm{g}(\epsilon, \phi, \mathbf{x})$

Algorithm

While not converged:

1. Sample data
2. Sample ϵ for each data point
3. Update via SGD

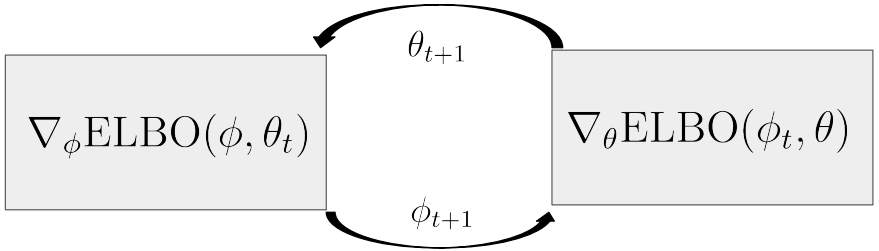


E.g.

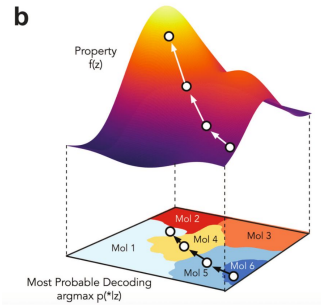
$$\epsilon \sim \mathcal{N}(0, \mathbf{I})$$

$$(\boldsymbol{\mu}, \log \boldsymbol{\sigma}) = \text{EncoderNeuralNet}_{\boldsymbol{\phi}}(\mathbf{x})$$

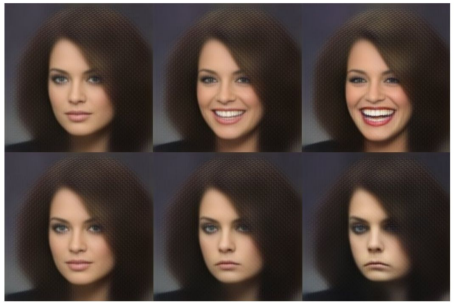
$$\mathrm{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$$



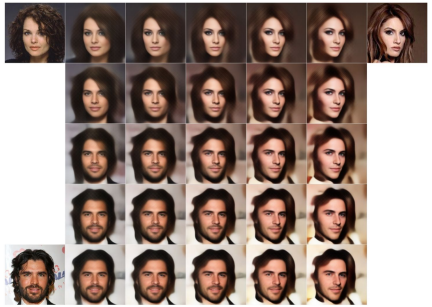
Applications: Representation Learning



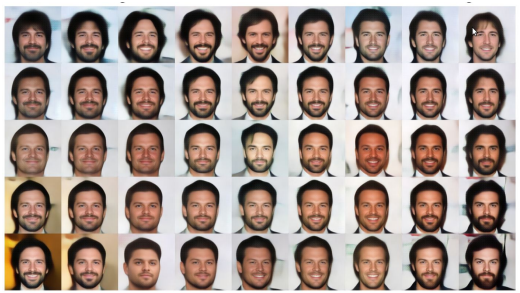
Discovery



Attributes



Analogies



Manifold Traversal

Beyond Likelihood: Information Maximization

Want to find latent Z that maximizes mutual information $\mathrm{I}(\mathrm{X};\ \mathrm{Z})$

$$\mathrm{I}(\mathrm{X};\ \mathrm{Z}) = \mathrm{H}(\mathrm{Z})\ \text{-}\ \mathrm{H}(\mathrm{Z} \mid X) = H(X) - H(X \mid Z)$$

Beyond Likelihood: Information Maximization

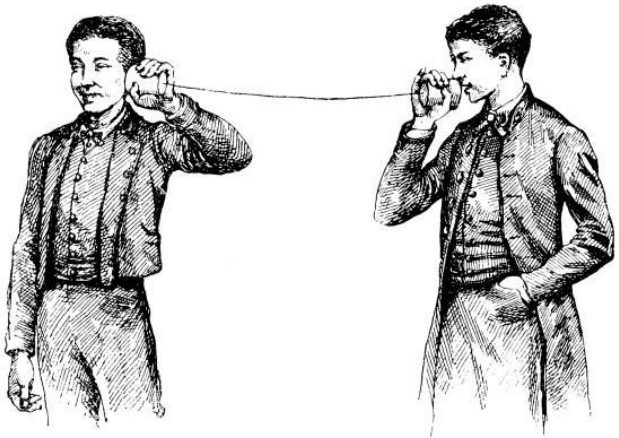
Want to find latent Z that maximizes mutual information $\mathrm{I}(\mathrm{X};\ \mathrm{Z})$

$$\mathrm{I}(\mathrm{X};\ \mathrm{Z}) = \mathrm{H}(\mathrm{Z}) - \mathrm{H}(\mathrm{Z} \mid X) = H(X) - H(X \mid Z)$$

Intuition:

You send signal **x** over a noisy wire, and receiver sees **y**. How much information does **y** tell you about **x**?

- No noise → all the bits in **x**.
- No signal → 0 bits.



Beyond Likelihood: Information Maximization

Want to find latent Z that maximizes mutual information $\mathrm{I}(\mathrm{X};\ \mathrm{Z})$

$$\mathrm{I}(\mathrm{X};\ \mathrm{Z}) = \mathrm{H}(\mathrm{Z}) - \mathrm{H}(\mathrm{Z} \mid X) = H(X) - H(X \mid Z)$$

$$-\sum p(\mathbf{x}, \mathbf{z}) \log p(\mathbf{z} \mid \mathbf{x})$$

$$-\sum p(\mathbf{x}) \log p(\mathbf{x})$$

Beyond Likelihood: Information Maximization

Want to find latent Z that maximizes mutual information $\mathrm{I}(\mathrm{X};\ \mathrm{Z})$

$$\mathrm{I}(\mathrm{X};\ \mathrm{Z}) = \mathrm{H}(\mathrm{Z}) - \mathrm{H}(\mathrm{Z} \mid X) = H(X) - H(X \mid Z)$$

$$-\sum p(\mathbf{x}, \mathbf{z}) \log p(\mathbf{z} \mid \mathbf{x})$$

$$-\sum p(\mathbf{x}) \log p(\mathbf{x})$$

Use **variational** approximation

Beyond Likelihood: Information Maximization

Formulate a **variational lower bound MI** (should feel familiar)

$$\mathrm{KL}(\mathrm{p}(\mathrm{z} \mid \mathbf{x}) \mid\mid q(\mathbf{z} \mid \mathbf{x}) \mid p(\mathbf{x})) \geq 0$$

Beyond Likelihood: Information Maximization

Formulate a **variational lower bound MI** (should feel familiar)

$$\mathrm{KL}(\mathrm{p}(\mathrm{z} \mid \mathbf{x}) \mid\mid q(\mathbf{z} \mid \mathbf{x}) \mid p(\mathbf{x})) \geq 0$$

$$\Downarrow$$

$$\sum p(\mathbf{x}, \mathbf{z}) \log \frac{p(\mathbf{z}|\mathbf{x})}{q(\mathbf{z}|\mathbf{x})} = \sum p(\mathbf{z}) \sum p(\mathbf{x} \mid \mathbf{z}) \left[\log p(\mathbf{z} \mid \mathbf{x}) - \log q(\mathbf{z} \mid \mathbf{x})\right]$$

Beyond Likelihood: Information Maximization

Formulate a **variational lower bound MI** (should feel familiar)

$$\mathrm{KL}(\mathrm{p}(\mathrm{z} \mid \mathbf{x}) \mid\mid q(\mathbf{z} \mid \mathbf{x}) \mid p(\mathbf{x})) \geq 0$$

$$\Downarrow$$

$$\sum p(\mathbf{x}, \mathbf{z}) \log \frac{p(\mathbf{z}|\mathbf{x})}{q(\mathbf{z}|\mathbf{x})} = \sum p(\mathbf{z}) \sum p(\mathbf{x} \mid \mathbf{z}) \left[\log p(\mathbf{z} \mid \mathbf{x}) - \log q(\mathbf{z} \mid \mathbf{x})\right]$$

$$\boxed{\sum p(\mathbf{x}, \mathbf{z}) \log p(\mathbf{z} \mid \mathbf{x}) \geq \sum p(\mathbf{z}) \sum p(\mathbf{x} \mid \mathbf{z}) \log q(\mathbf{z} \mid \mathbf{x})}$$

Beyond Likelihood: Information Maximization

Formulate a **variational lower bound MI** (should feel familiar)

$$\boxed{\sum p(\mathbf{x}, \mathbf{z}) \log p(\mathbf{z} \mid \mathbf{x}) \geq \sum p(\mathbf{z}) \sum p(\mathbf{x} \mid \mathbf{z}) \log q(\mathbf{z} \mid \mathbf{x})}$$

$$\Downarrow$$

$$\mathrm{I}(\mathrm{X};\, \mathrm{Z}) \geq \underbrace{H(\mathbf{Z}) - \sum p(\mathbf{z}) \sum_{h} p_\theta(\mathbf{x} \mid \mathbf{z}) \log q_\phi(\mathbf{z} \mid \mathbf{x})}_{L(\theta,\phi)}$$

Application: Skill Learning

Observations / states: s_t

Actions: a_t

Skill *representation*: z

Policy: $\underbrace{\pi_\theta(a_t \mid s_t, z)}_{\textbf{a skill}}$

Discrete $(0, 1, \ldots)$ or **continuous** (vector), sampled from $p(z)$.

Note: defining a skill requires **both** a skill representation (z) and policy parameters (θ).

We are optimizing **mutual information.**

What skills emerge from this game?

Exploration:
skills should cover the state space.

Predictability:
can you guess what a skill will do?

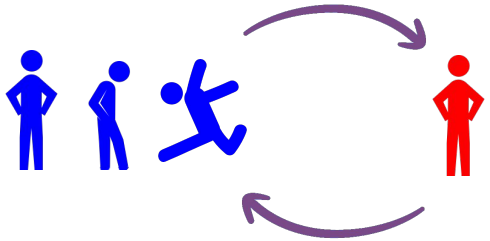
Learning skills by optimizing

$$I^{\pi}(z;\tau)$$
$$= \underline{\mathcal{H}[\tau]} - \underline{\mathcal{H}[\tau \mid z]}$$
$$= \mathcal{H}[z] - \mathcal{H}[z \mid \tau]$$

A constant, ignore this.

Can you predict which skill generated a trajectory?

Maximizing MI is a **cooperative** game with two players.



Policy / skills Discriminator

Learning skills by optimizing

$$I^{\pi}(z;\tau)$$
$$= \mathcal{H}[\tau] - \mathcal{H}[\tau \mid z]$$
$$= \mathcal{H}[z] - \mathcal{H}[z \mid \tau]$$
$$\geq \mathcal{H}[z] + \mathbb{E}_{p(z)\underline{p^{\pi}(s_t|z)}}[\log \underline{q(z \mid s_t)}]$$

Policy optimization

$$I^{\pi}(z;\tau)$$

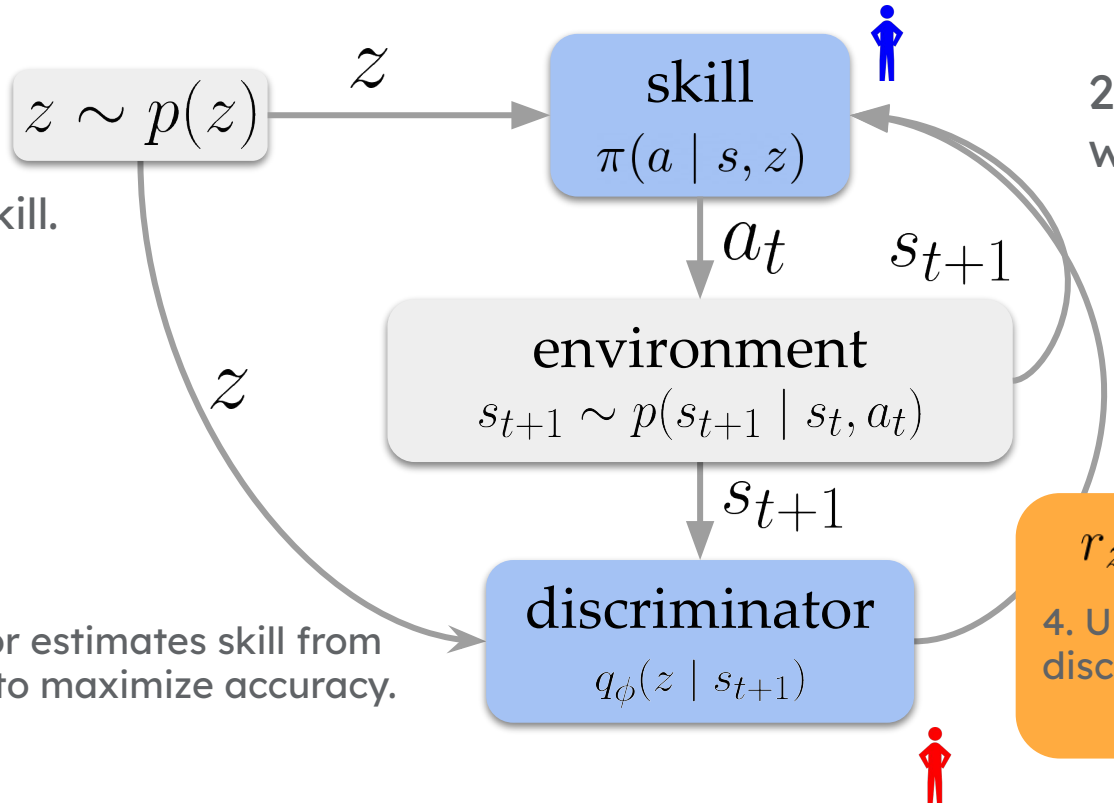
$$\geq \mathcal{H}[z] + \mathbb{E}_{p(z)\underline{p^{\pi}(s_t|z)}}[\log q(z \mid s_t)]$$

Apply RL to the reward

$$r_t = \log q(z \mid s_t)$$

$$\mathbb{E}_{p^{\pi}(s_f|z)}[\log q(z \mid s_t)] = (1-\gamma)\mathbb{E}_{\tau\sim\pi(\cdot|\cdot,z)}\left[\sum_t \gamma^t \underline{\log q(z \mid s_t)}\right]$$

Application: Skill Learning



1. Sample skill.

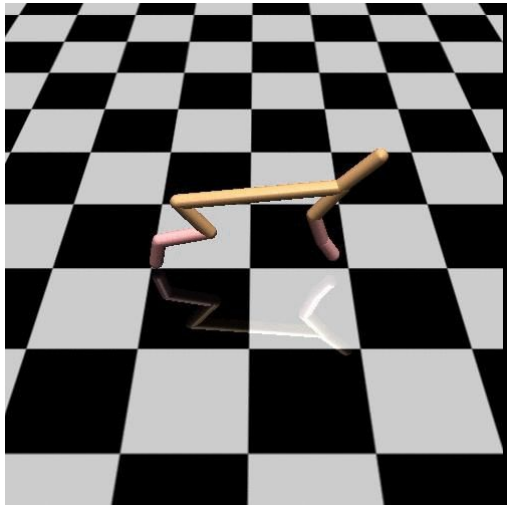
2. Collect one episode with this skill.

3. Discriminator estimates skill from state. Update to maximize accuracy.

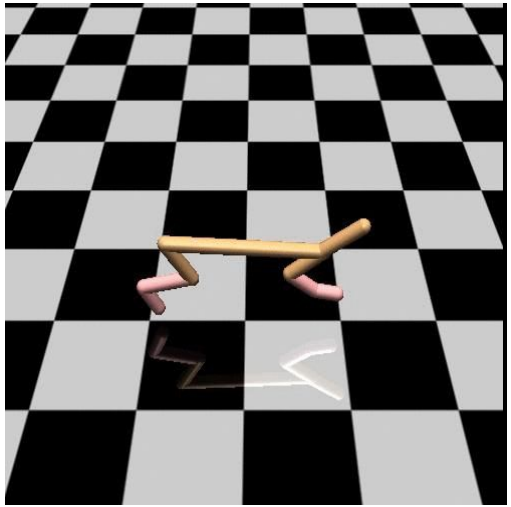
$$r_z(s) = \log q_\phi(z \mid s_{t+1})$$

4. Update skill to maximize discriminator accuracy.

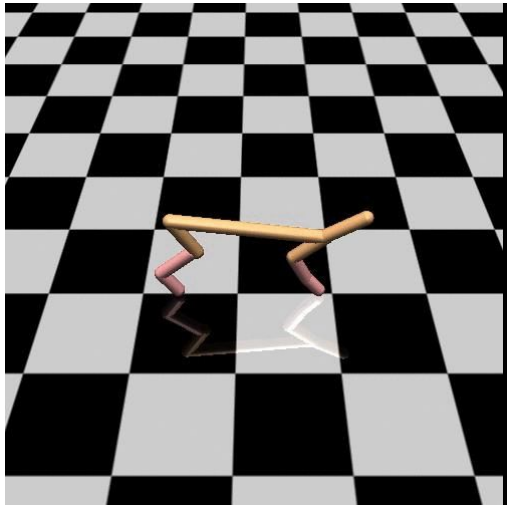
Does this work?



$\pi(a \mid s, z = 1)$



$\pi(a \mid s, z = 2)$



$\pi(a \mid s, z = 3)$

...

Thanks!